

FMCW Radar Cheatsheet

FMCW Radar Cheatsheet • v1.0, 2026.09.01 • <https://github.com/RadarML/radar-cheatsheet>

Parameters & Notation

c	299 792 458 m/s
B	swept bandwidth (MHz–GHz)
F_s	ADC sample rate (Ksps/Msps)
N	ADC samples per chirp
T_s	$= N/F_s$, chirp observation time
S	$= B/T_s = BF_s/N$ (MHz/ μ s)
f_0	chirp start frequency
λ	$= c/F$, carrier wavelength
T_c	chirp-to-chirp period (slow time)
N_d	chirps per frame
$N_d T_c$	coherent processing interval (CPI)
d	(virtual) antenna spacing
N_a	(virtual) receive antennas

TI Radars

- $B = \text{freq_slope} * \text{adc_samples} / \text{sample_rate}$
- $T_c = (\text{idle_time} + \text{ramp_end_time}) * \text{num_tx}$
(with TDM across Tx antennas)
- Sample order is (slow time, tx, rx, fast time)

Signal Model & Processing Chain

fast time $\rightarrow$ range, slow time $\rightarrow$ Doppler, antennas $\rightarrow$ angle

- Tx: $x_{Tx}(t) = A \exp(j(2\pi f_0 t + \pi \frac{B}{T} t^2))$
- Rx: same, with $t \rightarrow t - t_d$, $t_d = 2r/c$
- Dechirp (IF/beat): $x_{IF} = x_{Tx} x_{Rx}^*$

$$= A \exp(j(2\pi f_0 t_d + 2\pi S t_d t - \pi S t_d^2))$$

- Sample at $t = nT_s \Rightarrow$ cube $x_{IF}[n, n_d, n_a]$.

Range (fast-time FFT)

- Beat frequency: $f_b = S t_d = \frac{2Sr}{c} \Rightarrow r = \frac{f_b c}{2S}$
- Resolution: $\Delta r = \frac{c}{2B}$
- Max range: $R_{\max} = N \Delta r = \frac{F_s c}{2S}$
- Real-valued ADC (no I/Q): only $N/2$ usable bins, $f_{b,\max} = F_s/2$, so $R_{\max} = \frac{F_s c}{4S}$

Doppler (slow-time FFT)

- Round-trip phase: $\phi \approx 2\pi f_0 t_d = 4\pi r/\lambda$
- Per-chirp advance: $\Delta\phi_{dop} = \frac{4\pi v}{\lambda} T_c \Rightarrow v = \frac{\Delta\phi_{dop} \lambda}{4\pi T_c}$
- Doppler shift: $f_d = 2v/\lambda$
- Resolution: $\Delta v = \frac{\lambda}{2N_d T_c}$
- Max unambiguous: $|v| \leq v_{\max} = \frac{\lambda}{4T_c}$
(from $|f_d| \leq \frac{1}{2T_c}$, or $|\Delta\phi_{dop}| < \pi$)

FFT Bin $\rightarrow$ Frequency

For $X[k] = \sum_n x[n] e^{-j2\pi kn/N}$, the upper half of the index range is negative frequency:

$$f_k = \begin{cases} \frac{k}{N} F_s, & 0 \leq k \leq N/2 \\ \frac{k-N}{N} F_s, & N/2 < k \leq N-1 \end{cases}$$

- Range: $r \geq 0$ always $\Rightarrow$ **no fftshift**.
- Doppler (v) and angle ($\sin \theta$) are signed $\Rightarrow$ **must fftshift**.

Array Beamforming

One complex sample per antenna, per range-Doppler bin.

- Steering vector, element positions p_m :

$$[a(\theta)]_m = \exp(-j \frac{2\pi}{\lambda} p_m \sin \theta)$$

- Model (single source): $\mathbf{x}(t) = a(\theta)s(t) + \mathbf{n}(t)$
- Output: $y(t; \theta) = a^H(\theta)\mathbf{x}(t)$
- Sweep θ , plot $|y(t; \theta)|^2 \Rightarrow$ angular spectrum.

Angle FFT

- Across-antenna signal: $x[n] = A \exp(j \frac{2\pi}{\lambda} n d \sin \theta)$
- Peak at $\frac{k}{N_a} = \frac{d \sin \theta}{\lambda}$, so bin k of the FFT maps to

$$\sin \theta = \frac{k\lambda}{N_a d} \stackrel{d=\lambda/2}{=} \frac{2k}{N_a}$$

- Resolution: $\Delta(\sin \theta) = \frac{\lambda}{N_a d}$ (aperture $N_a d$)
- In angle: $\Delta\theta = \frac{\lambda}{N_a d \cos \theta}$; near broadside $\Delta\theta \approx \lambda/D$, $D = (N_a - 1)d$; diverges as $\theta \rightarrow \pm 90^\circ$.
- Bins are linear in $\sin \theta$, *not* in θ .
- Need $d \leq \lambda/2$, else grating lobes.
- 2D arrays: same idea, 2D DFT, per-axis N_a and d .

Zero-Pad / Crop / Decimate

Op	Resolution	Max / bins
Pad $N \rightarrow N' > N$	unchanged	max unchanged; more bins (interpolation only)
Crop $N \rightarrow N' < N$	coarsens $\propto N/N'$	fewer bins
Decimate $/D$	unchanged	max $/D \Rightarrow$ risk of aliasing

Per-axis specifics of decimation by D :

- Range: $F_s \rightarrow F_s/D$, $R_{\max}/D$ — low-pass filter first.
- Doppler: $T_c \rightarrow DT_c$, $v_{\max}/D$ — alias unless $|v|$ is known small.
- Angle: $d \rightarrow Dd$; aperture kept, but $Dd > \lambda/2$ gives grating lobes.

1 Signal Model

Definition 1.1 (Transmitted Chirp). $x_{Tx}(t) = A \exp(j(2\pi f_0 t + \pi \frac{B}{T} t^2))$, where f_0 is the chirp start frequency and T is the chirp duration (often $T \approx T_s$).

This follows from integrating the instantaneous frequency $f(t) = f_0 + \frac{B}{T}t$ (a linear ramp from f_0 to $f_0 + B$ over duration T): $\phi(t) = 2\pi \int_0^t f(t') dt' = 2\pi f_0 t + \pi \frac{B}{T} t^2$.

Definition 1.2 (Received Chirp). For a target at range r with round-trip delay $t_d = 2r/c$, $x_{Rx}(t) = A \exp(j(2\pi f_0(t - t_d) + \pi \frac{B}{T}(t - t_d)^2))$.

Definition 1.3 (IF / Beat Signal). Dechirp mixing multiplies the received signal by the conjugate of the transmitted signal:

$$x_{IF}(t) = x_{Tx}(t) x_{Rx}^*(t) = A \exp\left(j\left(2\pi f_0 t_d + 2\pi \frac{B}{T} t_d t - \pi \frac{B}{T} t_d^2\right)\right),$$

sampled at $t = nT_s$ to give $x_{IF}[n]$.

2 Range Processing

Definition 2.1 (Beat Frequency). $f_b = \frac{B}{T} t_d = S t_d = \frac{2Sr}{c}$, so $r = \frac{f_b c}{2S}$.

The range FFT recovers f_b per target as a spectral peak. With x_{IF} complex/IQ, f_b is unambiguous up to F_s , matching $R_{\max} = F_s c / (2S)$ below.

Remark 2.1 (Real-Valued ADC Sampling). If the ADC instead samples a real-valued IF signal (no I/Q mixer), $x_{IF}[n]$ is real, so its N -point DFT is conjugate-symmetric and only the $N/2$ bins from 0 to $F_s/2$ are usable (the upper half mirrors the lower half rather than encoding new information). Nyquist then caps the beat frequency at $f_{b,\max} = F_s/2$, halving the usable range bins and giving $R_{\max} = \frac{F_s c}{4S}$.

Remark 2.2. We will assume complex/IQ sampling unless stated otherwise.

Definition 2.2 (Range FFT). $X[r] = \sum_{n=0}^{N-1} x_{IF}[n] \exp(-j2\pi \frac{rn}{N})$.

Definition 2.3 (Bandwidth). Radar bandwidth (frequency) B available to be swept during the chirp period; typically MHz or GHz.

Remark 2.3 (TI Implementation). On a TI radar, this is `freq_slope * adc_samples / sample_rate`.

Definition 2.4 (Speed of Light). $c = 299792458 \text{ m/s}$.

Definition 2.5 (Sample rate). Sampling frequency F_s of the radar ADC.

Remark 2.4 (Units). This is typically specified in KHz (Ksps) or MHz (Msps).

Definition 2.6 (Number of samples). Number of ADC samples N per chirp; this leads to a total observation time of $T_s = N/F_s$.

Definition 2.7 (Slope). Slope $S = B/T_s = BF_s/N$ of the chirp waveform; typically MHz/ μs .

Definition 2.8 (Range Resolution). $\Delta r = \frac{c}{2B}$.

Suppose a given object is at range r . This leads to time delay $\tau = 2r/c$, so $\Delta r = c\Delta\tau/2$.

When taking a DFT over the chirp, the frequency resolution is $1/T_s$. Since the frequency resolution corresponding to a time resolution is S , the time delay resolution is $\Delta\tau = \frac{1/T_s}{S} = \frac{1}{B}$. This leads to $\Delta r = c\Delta\tau/2 = \frac{c}{2B}$.

Definition 2.9 (Maximum Range). $R_{\max} = N \frac{c}{2B} = \frac{F_s c}{2S}$.

This can be seen by combining the range resolution with the number of range bins, which is equal to the DFT window size (N).

Remark 2.5 (Padding). Zero-padding the N ADC samples to $N' > N$ before the range FFT increases the number of range bins via spectral interpolation, but does not change the true range resolution $\Delta r = c/2B$ (set by the swept bandwidth B) or the true maximum unambiguous range (set by the sample rate F_s and slope S).

Remark 2.6 (Cropping). Using fewer than N of the acquired samples shortens the effective bandwidth B and observation window T_s , coarsening the range resolution Δr proportionally (since the frequency resolution of the shorter DFT is $1/T'_s$ for the cropped window $T'_s < T_s$), and reduces the number of range bins accordingly.

Remark 2.7 (Decimation). Keeping every D -th ADC sample reduces the effective sample rate to F_s/D ; the observation window T_s is unaffected, so the range resolution $\Delta r = c/2B$ is unchanged, but the maximum unambiguous range $R_{\max} = F_s c/(2S)$ shrinks by a factor of D , risking range aliasing unless the signal is properly low-pass filtered first.

3 Doppler Processing

Definition 3.1 (Wavelength). $\lambda = c/F$.

Remark 3.1 (Narrow-Band Approximation). If using a narrow-band radar, λ can be approximated as a constant with respect to the center frequency of the chirp.

Definition 3.2 (Chirp Rate). *Time between chirps* T_c .

Remark 3.2 (TI Implementation). On a TI radar, if using time-division multiplexing across antennas, this is `(idle_time + ramp_end_time)* num_tx`.

Definition 3.3 (Round-Trip Phase). $\phi \approx 2\pi f_0 t_d = 2\pi f_0 \cdot \frac{2r}{c} \approx \frac{4\pi r}{\lambda}$.

Definition 3.4 (Doppler Phase Evolution). *Over slow time* $t_{\text{slow}} = nT_c$ ($n = 1, 2, \dots, N_d$ chirps), with $R(t_{\text{slow}}) = R_0 + vt_{\text{slow}}$:

$$\phi_{\text{dop}}(t_{\text{slow}}) = \frac{4\pi R(t_{\text{slow}})}{\lambda} = \frac{4\pi R_0}{\lambda} + \frac{4\pi v}{\lambda} t_{\text{slow}}.$$

Definition 3.5 (Velocity from Phase Difference). *The phase advances by* $\Delta\phi_{\text{dop}} = \phi_{\text{dop}}[n] - \phi_{\text{dop}}[n-1] = \frac{4\pi v}{\lambda} T_c$ *per chirp, giving the two-chirp velocity estimate* $v = \frac{\Delta\phi_{\text{dop}} \lambda}{4\pi T_c}$.

Definition 3.6 (Doppler Resolution). $\Delta v = \frac{\lambda}{2N_d T_c}$.

Suppose a given object has radial velocity v . The two-way Doppler shift is $f_d = 2v/\lambda$, so $v = f_d \lambda/2$ and $\Delta v = \Delta f_d \lambda/2$.

When taking a DFT over the N_d chirps of a frame (slow time), the frequency resolution is $1/(N_d T_c)$. This is exactly the Doppler frequency resolution Δf_d , so $\Delta v = \Delta f_d \lambda/2 = \frac{\lambda}{2N_d T_c}$.

Definition 3.7 (Maximum Doppler Velocity). $v_{\max} = \frac{\lambda}{4T_c}$.

Chirps are sampled once every T_c (slow-time sampling rate $1/T_c$), so by the Nyquist criterion the unambiguous Doppler frequency is limited to $|f_d| \leq \frac{1}{2T_c}$. Since $v = f_d \lambda/2$, this gives $v_{\max} = \frac{1}{2T_c} \cdot \frac{\lambda}{2} = \frac{\lambda}{4T_c}$.

Remark 3.3 (Alternative Max-Velocity Derivation). Equivalently, from the phase-difference view above, unambiguous phase requires $|\Delta\phi_{\text{dop}}| < \pi$, i.e. $\frac{4\pi v}{\lambda} T_c < \pi$, giving the same $v_{\max} = \frac{\lambda}{4T_c}$.

Remark 3.4 (FFT Bin Sign Interpretation). For a DFT

$$X[k] = \sum_n x[n] \exp(-j2\pi kn/N),$$

bin $k = N - 1$ is identical to bin $k = -1$ since

$$\exp(-j2\pi(N-1)n/N) = \exp(j2\pi n/N) \exp(-j2\pi n) = \exp(j2\pi n/N),$$

so bins in the upper half of the index range represent negative frequencies:

$$f_k = \begin{cases} \frac{k}{N} F_s, & 0 \leq k \leq N/2 \\ \frac{k-N}{N} F_s, & N/2 < k \leq N-1. \end{cases}$$

Because measured range is always non-negative, the Range FFT does not require any `fftshift`. But Doppler velocity (and, in the next section, $\sin \theta$) can be negative, so Doppler and Angle FFT outputs must be reinterpreted (`fftshift`-ed) using the formula above to recover signed velocity/angle.

Remark 3.5 (Padding). Zero-padding the N_d chirps to $N'_d > N_d$ before the Doppler FFT increases the number of Doppler bins via spectral interpolation, but does not change the true Doppler resolution $\Delta v = \lambda/(2N_d T_c)$ (set by the number of chirps actually collected) or the maximum unambiguous velocity.

Remark 3.6 (Cropping). Using fewer than N_d of the collected chirps shortens the coherent processing interval $N_d T_c$, coarsening the Doppler resolution $\Delta v = \lambda/(2N_d T_c)$ proportionally, and reduces the number of Doppler bins accordingly.

Remark 3.7 (Decimation). Keeping every D -th chirp increases the effective chirp period to DT_c ; since the total coherent processing interval $N_d T_c$ is unchanged, the Doppler resolution $\Delta v = \lambda/(2N_d T_c)$ is preserved, but the maximum unambiguous velocity $v_{\max} = \lambda/(4T_c)$ shrinks by a factor of D , risking Doppler aliasing unless the target's true velocity is known to lie within the new (narrower) unambiguous range.

4 Range-Doppler Geometry

Remark 4.1. This section is useful for radar *rendering*, but generally not relevant to perception. For a deep dive into how this can be used, as well as an expanded derivation (Appendix A.5), see *Implicit Doppler Tomography for Radar Novel View Synthesis*, Huang et. al. (CVPR '24).

4.1 Inputs

Definition 4.1 (Velocity). We denote the velocity direction as $\mathbf{v}$, where $\|\mathbf{v}\|_2 = 1$.

Definition 4.2 (Speed). Denote the magnitude component as s , where the full velocity is $\mathbf{v}s$.

Definition 4.3 (Measurements: Distance (radius), Doppler). (r, d) bins; assuming the sensor position is $\mathbf{0}$, for positions $\mathbf{a}$, this constrains measurements to $\|\mathbf{a}\|_2 = r$ and $\frac{\mathbf{a}^T \mathbf{v}s}{\|\mathbf{a}\|_2} = \frac{\mathbf{a}^T \mathbf{v}s}{r} = d$.

The target integration is described by

$$y(r, d; \mathbf{x}_{\text{sensor}}, \mathbf{v}) = \int_{\|\mathbf{t}\|_2=r, \frac{\mathbf{t}^T \mathbf{v}}{\|\mathbf{t}\|_2}=d} \sigma(\mathbf{x}_{\text{sensor}} + \mathbf{t}) d\mathbf{t}$$

4.2 Integration Circle

The constraints are a sphere and a plane, which should constrain measurements to a circle. The center of the circle is $\mathbf{c} = \mathbf{v} \frac{rd}{s}$, since this is in the direction of the normal vector $\mathbf{v}$ and satisfies

$$\frac{(\mathbf{v}(rd/s))^T \mathbf{v}s}{r} = \frac{\mathbf{v}^T \mathbf{v}s(rd/s)}{r} = d.$$

The radius of this circle is

$$r' = \sqrt{r - \left\| \mathbf{v} \frac{rd}{s} \right\|^2} = \sqrt{r - \left(\frac{rd}{s} \right)^2} = r \sqrt{1 - \left(\frac{d}{s} \right)^2}.$$

Let $\mathbf{p}$ and $\mathbf{q}$ form an orthonormal basis with $\mathbf{v}$ (i.e. via Gram-Schmidt). Then, the circle can be parameterized by angle ψ as

$$\begin{aligned}\mathbf{a} &= r'\mathbf{p}\cos(\psi) + r'\mathbf{q}\sin(\psi) + \mathbf{v}\frac{rd}{s} \\ &= r\left(\sqrt{1-(d/s)^2}(\mathbf{p}\cos(\psi) + \mathbf{q}\sin(\psi)) + \mathbf{v}\frac{d}{s}\right).\end{aligned}$$

5 MIMO and the Virtual Array

Angle estimation requires many spatially separated measurements, but physical elements are costly. Since only the *sum* of the transmit and receive path lengths appears in the measured phase, transmitting from several separated elements synthesizes a larger effective array.

Definition 5.1 (Transmit and Receive Arrays). N_{tx} transmit elements at $p_0^{tx}, \dots, p_{N_{tx}-1}^{tx}$ and N_{rx} receive elements at $p_0^{rx}, \dots, p_{N_{rx}-1}^{rx}$, along a common 1-D array axis.

Definition 5.2 (Virtual Array). For a target at angle θ , pair (i, j) measures round-trip phase

$$\phi_{ij} = \frac{2\pi}{\lambda} (p_i^{tx} + p_j^{rx}) \sin \theta,$$

identical to a single physical element at $p_i^{tx} + p_j^{rx}$. The virtual array is the set of these sums,

$$\{p_m\} = \{p_i^{tx} + p_j^{rx} : 0 \leq i < N_{tx}, 0 \leq j < N_{rx}\}, \quad N_a = N_{tx}N_{rx}.$$

Against a reference element at the origin, the extra path length is $p_i^{tx} \sin \theta$ on transmit and $p_j^{rx} \sin \theta$ on receive; the round trip accumulates both.

Remark 5.1 (Uniform Virtual Array (ULA)). Receive elements spaced d and transmit elements spaced $N_{rx}d$ apart make the sums $p_i^{tx} + p_j^{rx}$ tile a uniform linear array of $N_{tx}N_{rx}$ elements at spacing d , with no gaps or repeats.

Remark 5.2 (Resolution Gain). The angular resolution $\Delta(\sin \theta) = \frac{\lambda}{N_a d}$ is set by the *virtual* aperture $N_a d$, so MIMO gains a factor of N_{tx} over the receive array alone while costing only $N_{tx} + N_{rx}$ physical elements.

Definition 5.3 (TDM-MIMO). Exactly one transmit element is active per chirp, cycling through the N_{tx} elements in order, so each transmitter's contribution is separated by its time slot. One pass through all N_{tx} is a loop, and a frame is N_d loops.

Definition 5.4 (Ramp Repetition Interval). Time T_r between consecutive chirps of any transmitter, including idle time. This differs from the chirp duration T , which covers only the active sweep.

Remark 5.3 (Alternatives to TDM). Doppler-division multiplexing (DDM) instead transmits from all elements simultaneously, separating them by (coded) Doppler offset. This avoids the N_{tx} -fold reduction in $v_{\max}$, at the cost of additional separation processing.

6 Array Beamforming

After range-Doppler processing, a given range-Doppler bin contains one complex sample per receive antenna. To estimate the angle θ of the target from these per-antenna phases, we look at the array response across antennas.

Definition 6.1 (Steering Vector). For an array with element positions $p_0, p_1, \dots, p_{N_a-1}$ along a 1-D array axis, the array vector is

$$\mathbf{a}(\theta) = \begin{bmatrix} \exp\left(-j\frac{2\pi}{\lambda}p_0 \sin \theta\right) \\ \exp\left(-j\frac{2\pi}{\lambda}p_1 \sin \theta\right) \\ \vdots \\ \exp\left(-j\frac{2\pi}{\lambda}p_{N_a-1} \sin \theta\right) \end{bmatrix}$$

for a target at angle θ .

Definition 6.2 (Array Data Model). Assuming a single target/source, $\mathbf{x}(t) = a(\theta)s(t) + \mathbf{n}(t)$, where $s(t)$ is the per-antenna baseband signal and $\mathbf{n}(t)$ is noise.

Definition 6.3 (Beamformed Output). $y(t; \theta) = a^H(\theta) \mathbf{x}(t)$.

Remark 6.1 (Angular Spectrum). Steering (conjugate-matching) the array response to angle θ coherently combines the signal from that direction. Accordingly, sweeping θ (i.e., with arbitrary sampling resolution) and evaluating $|y(t; \theta)|^2$ recovers the angular spectrum for the given array geometry.

7 Angular FFT Processing

When the array is uniform, $p_m = md$, and sweeping θ in the beamformer above becomes equivalent to a spatial DFT across antennas, computable efficiently via FFT.

Definition 7.1 (Antenna Spacing). Spacing d between elements of the (virtual) receive array.

Remark 7.1 (Typical Value). Radars typically set to $d = \lambda/2$ to avoid spatial aliasing.

Definition 7.2 (Number of Antennas). Number of (virtual) receive antenna elements N_a in the array.

Let θ be the angle to a target, measured from broadside (i.e., the array normal) of a linear array with N_a elements and spacing d . The extra path length to antenna n relative to antenna 0 is $nd \sin \theta$, giving a phase offset of $\frac{2\pi}{\lambda} nd \sin \theta$. The spatial signal across antennas is therefore

$$x[n] = A \exp \left(j \frac{2\pi}{\lambda} nd \sin \theta \right), \quad n = 0, \dots, N_a - 1.$$

Taking an N_a -point DFT over this spatial signal (i.e., across antennas for a fixed range-Doppler bin) yields the angular spectrum

$$X[k] = \sum_{n=0}^{N_a-1} x[n] \exp \left(-j \frac{2\pi}{N_a} kn \right),$$

which is maximized (constructive interference) when $\frac{k}{N_a} = \frac{d \sin \theta}{\lambda}$.

Definition 7.3 (Angular Bin Mapping). Bin k of the angular spectrum corresponds to $\sin \theta = \frac{k\lambda}{N_a d}$; with the typical spacing $d = \lambda/2$, this is $\sin \theta = \frac{2k}{N_a}$.

Remark 7.2 (Non-linearity). Unlike range and Doppler, the mapping from bin index to angle is non-linear in θ (linear only in $\sin \theta$), so angular bins are spaced increasingly far apart in θ towards $\pm 90^\circ$.

Remark 7.3 (2D Arrays). For 2D arrays, the same principle applies in both dimensions, with the angular spectrum being a 2D DFT over the 2D array, and the angles corresponding to each bin being determined by the number of antennas and spacing in each dimension.

Definition 7.4 (Angular Resolution). $\Delta(\sin \theta) = \frac{\lambda}{N_a d}$.

As with Doppler, the DFT over the N_a -antenna aperture has a (normalized spatial) frequency resolution of $1/N_a$ in $\frac{d \sin \theta}{\lambda}$, so

$$\Delta \left(\frac{d \sin \theta}{\lambda} \right) = \frac{1}{N_a} \implies \Delta(\sin \theta) = \frac{\lambda}{N_a d}.$$

In other words, given a physical aperture $N_a d$, two targets closer than $\Delta(\sin \theta)$ apart cannot be separated into distinct peaks.

Remark 7.4 (Resolution in Angle). Resolution in θ coarsens away from broadside: since $\frac{d(\sin \theta)}{d\theta} = \cos \theta$,

$$\Delta \theta = \frac{\Delta(\sin \theta)}{\cos \theta} = \frac{\lambda}{N_a d \cos \theta}.$$

Near broadside ($\theta \approx 0$), this simplifies to $\Delta \theta \approx \lambda/D$, where $D = (N_a - 1)d$ is the array aperture. However, as $\theta \rightarrow \pm 90^\circ$, $\cos \theta \rightarrow 0$ and $\Delta \theta$ diverges.

Remark 7.5 (Padding). Zero-padding the N_a antenna samples to $N'_a > N_a$ before the angular FFT increases the number of angular bins via spectral interpolation, but does not change the true angular resolution $\Delta(\sin \theta) = \lambda/(N_a d)$ (set by the physical aperture $N_a d$).

Remark 7.6 (Cropping). Using fewer than N_a of the array elements shortens the effective aperture, coarsening the angular resolution $\Delta(\sin \theta)$ proportionally, and reduces the number of angular bins accordingly.

Remark 7.7 (Decimation). Keeping every D -th antenna element over the same physical aperture increases the effective spacing to Dd ; since the aperture $N_a d$ is unchanged, the angular resolution $\Delta(\sin \theta) = \lambda/(N_a d)$ is preserved, but exceeding the $d = \lambda/2$ spatial Nyquist limit introduces grating lobes (spatial aliasing), causing ambiguous angle estimates.